

# Dipayan Sarkar

Department of Bioinformatics, University of North Bengal

Siliguri - 734013, West Bengal, India.

☎ +91 9735490241 • ✉ dipayanskr@gmail.com • 🌐 dipayansarkar.com  
🐙 github.com/dipayanskr • 📄 Google Scholar • 🆔 0000-0002-0765-5689  
✉ rs\_dipayans@nbu.ac.in • 🤖 huggingface.co/dipayanskr

## Research Summary

PhD researcher in bioinformatics and computational biology developing deep learning methods for protein sequence modeling. Work spans sequence-based protein-protein interaction prediction using attention-based architectures with transfer learning from pretrained protein language models, generative modeling of protein binders through autoregressive and diffusion-based approaches. Also builds web platforms that make trained models usable for inference and interaction-network visualization.

## Education

**University of North Bengal, Department of Bioinformatics.**

**India**

*Ph.D. in Bioinformatics*

*Expected Submission 2027*

Supervisor: Dr. Chiranjib Sarkar, Assistant Professor, Department of Bioinformatics.

Focus: Framework Development for Predicting Protein-Protein Interaction Networks Using Deep Learning Algorithms on Protein Sequences.

**University of North Bengal, Department of Botany.**

**India**

*M.Sc. in Botany, 77.25% - First Class*

*2021*

Specialization: Biochemistry.

Dissertation: "A review on the responses of aquatic macrophytes to phenolic compounds with special reference to their tolerance strategies and future implications."

Supervisor: Dr. Swarnendu Roy, Assistant Professor, Department of Botany.

**Ananda Chandra College, University of North Bengal**

**India**

*B.Sc. (Hons.) in Botany, 63.50% - First Class*

*2019*

## Research Experience

**Department of Bioinformatics, University of North Bengal**

**Siliguri, India**

*PhD Research Scholar*

*2022–Present*

Department of Bioinformatics

- Deep learning architectures for sequence-based protein-protein interaction prediction using Attention-based hybrid architecture with transfer learning.
- Sequence based generative modeling of protein binder generation using autoregressive approaches.
- Development of web-based platforms for model inference and protein interaction network visualization

## Publications

**2025: Sarkar, D., & Sarkar, C. AttnSeq-PPI: Enhancing protein-protein interaction network prediction using transfer learning-driven hybrid attention. *Biochimica et Biophysica Acta (BBA)—Proteins and Proteomics*, 1874(1), 141102. doi:10.1016/j.bbapap.2025.141102**

**2026: Sarkar, D., & Sarkar, C. ARACoFusion: Uncertainty-aware calibrated deep learning for protein-protein interaction network prediction in *Arabidopsis thaliana*. *bioRxiv* preprint. doi:10.64898/2026.05.22.727120**

**2026: Sarkar, D., & Sarkar, C. MoE-Bind: Guiding De Novo Protein Binder Generation with Sparse Experts. *bioRxiv* preprint. doi:10.64898/2026.06.13.732043**

**2026: Sarkar, D., Bardhan, K., & Sarkar, C. Deep-Interact Studio: An Interactive Deep Learning Model Building Platform for Biomolecular Interaction Prediction. *bioRxiv* preprint. doi:10.64898/2026.07.02.736034**

## Research Software Developed

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### Web Servers.....

**Deep-Interact Studio:** Interactive web platform for building, training, and comparing custom deep learning models for protein–protein, drug–target, RNA–protein, and protein–DNA interaction prediction, with integrated interpretability.

[deepinteract.compbiosysnbu.in](http://deepinteract.compbiosysnbu.in)

**AttnSeq-PPI:** Deep learning-based protein–protein interaction prediction platform.

[attnseqppi.compbiosysnbu.in](http://attnseqppi.compbiosysnbu.in)

### Packages.....

**EGRNi:** R package for ensemble gene regulatory network inference, combining multiple inference methods to improve reconstruction accuracy of gene regulatory networks.

Sarkar, C., **Sarkar, D.**, Parsad, R., & Mishra, D. *CRAN*, v0.1.6

[cran.r-project.org/package=EGRNi](https://cran.r-project.org/package=EGRNi)

## Skills

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**Programming:** Python (PyTorch), R, C

**Deep Learning:** Transformers, attention mechanisms, transfer learning, pre-training and finetuning.

**Generative AI:** Autoregressive generative models for de novo protein and protein binder design.

**Scalable Training:** GPU-based training, parallel training on HPC environments

**Bioinformatics:** Biological sequence analysis, protein-protein interaction and biomolecular network prediction.

**Scientific Computing:** NumPy, Pandas, Matplotlib, Seaborn.

**Deployment:** Nginx, Docker, Flask, FastAPI, Linux server deployment, model-backed web applications.

**Tooling:** Git, GitHub Actions (CI), uv, Conda.

## Talks & Presentations

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**2025: A Webtool for Transfer Learning Based Protein-Protein Interaction Network Prediction Using Hybrid Attention** - Oral presentation, *Anusandhan 2025* (SARANSH, Wiley-sponsored), Bose Institute, Kolkata, India, 7 March 2025.

## Awards

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**2021: CSIR-UGC NET with JRF**, Life Sciences - June 2021

*All India Rank 216 | Junior Research Fellowship, CSIR-HRDG, Government of India*

**2022: Qualified GATE** (Graduate Aptitude Test in Engineering) - Life Sciences (XL)

## Grants & Computing Resources

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**2025: IndiaAI Compute Initiative** - Awarded GPU cloud compute (NVIDIA L4 GPU) under the IndiaAI Mission for the project *Deep learning-based framework for protein-protein interaction network prediction* (Project ID: P1-S2025070964, Sep 2025 - Aug 2026).

**2025: Google TRC (TPU Research Cloud) Award** - Awarded access to GCP - TPUs (v4, v5e, v6e) for machine learning research.